

Bryce Hanna

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Education

Georgia Institute of Technology

Atlanta, GA

Bachelor of Science in Computer Science

August 2023 - May 2027

Concentration: Systems & Architecture and Theory

GPA: 4.0/4.0

Relevant Coursework: Operating Systems, Digital Design, Compilers

Skills

Languages: Python, C/C++, TypeScript, Go, Bash, C#, Java, Lua, JavaScript, Rust, SQL, Assembly

Technologies: Linux Systems Programming, pthreads, ZeroMQ, FreeRTOS, protobuf, SDR, OpenCV, I2C/UART, React, SQLite

Developer Tools: Git, Bash/Zsh/fish, Docker, Grafana, Make/CMake, ssh/scp, tmux, Gitlab CI, Conda, Unity, (Neo)vim

Experience

Stripe

May 2026 - August 2026

Incoming Software Engineering Intern

South San Francisco, CA

Verkada

January 2026 - April 2026

Software Engineering Intern | Camera Core Systems Team

San Mateo, CA

- Owned end-to-end development of the camera light accessory firmware, architecting a C++ runt service while coordinating with hardware and product teams to investigate, design, debug, and iterate on tens of features for the new product
- Reduced accessory firmware failures by 50%, iterating on UART stability and task scheduling on FreeRTOS
- Slashed battery firmware upgrade time by 80% through optimization of the RS485 UART connection and scheduling timing
- Integrated I2C-based Gauge ICs with ZephyrOS in C to automate the collection and Grafana visualization of critical battery metrics for a fleet of 10,000+ cameras contributing to \$6.5M in revenue

Georgia Tech Research Institute

May 2025 - August 2025

Software Engineering Intern | Spectrum Warfare Operations and Research Division

Smyrna, GA

- Redesigned the Threat Sensor Emulator (TSE) radar simulator with threading and protobuf API with 1 GB/s throughput
- Increased performance of CFAR detection algorithm by 30x using vectorized numpy operations and numba JIT compiling
- Architected a GPS activated deception program on a UHD Software Defined Radio spanning 56 MHz of bandwidth

Educational Computer Science YouTube Channel (@BRicey)

March 2020 - Present

Technical Content Creator & Educator

- Script, record, and edit programming tutorials on advanced concepts in Roblox Studio contributing to 45 million game visits
- Educate an audience of 20,000+ subscribers with over 2 million total views on essential CS concepts
- Analyze retention data to increase content interactions to reach an annual revenue of over \$1,500

Research & Publications

Intuitive Navigation Assistance from an Exoskeleton | Python, GPS, Linux

March 2025 - December 2025

- Architected a multithreaded Python system on a raspberry Pi to navigate user through exoskeleton movements
- Orchestrated Camera, IMU, GPS, and high-powered leg-mounted motors using RedisDB, web sockets, and gpsd
- Extensively tested GPS and camera integration to reduce interference and errors by 50% using a custom web dashboard

XR Treatment for Age-Related Macular Degeneration (AMD) | Unity, C#, React

August 2024 - May 2025

- Leveraged Meta XR SDKs to track eye movement in Unity to diagnose and treat AMD across 3+ retinal evaluations
- Engineered a full-stack dashboard using React, Apache, and SQLite to visualize test results for doctors at Emory University
- **Publication:** "Asymmetric VR Design to Support Connected Care for Older Adults with AMD" submitted to DRS, 2026.

Leadership

School of Computing Instruction at Georgia Tech

July 2024 - Present

CS 2110 Undergraduate Teaching Assistant

Atlanta, GA

- Instruct a lab of 50+ students on binary computation, circuit logic, computer architecture, assembly, and C programming
- Assist hundreds of students in office hours and forums to facilitate mastery through personalized teaching techniques
- Create and debug teaching tools such as circuit/assembly simulators and autograders to aid student learning